SMDM PROJECT REPORT

Data Science and Business Analytics

Case Study 1: Enhancing Marketing Campaign Efficiency for Austo Motor Company

Case Study 2: Optimizing Credit Card Strategy for Revenue Generation in GODIGT Bank

Great Learning

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Case Study 1: Enhancing Marketing Campaign Efficiency for Austo Motor Company

Problem 1

1.1 What is the important technical information about the dataset that a database administrator would be interested in?

- **Shape of the dataset:** The dataset contains 1581 observations and 14 variables.
- **List of datatypes:** The dataset contains 1 float type, 5 Integer type and 8 object type

Float type: Partner_salary

Integer type: Age, No of Dependents, Salary, Total salary, and Price.

Object Type: Gender, Profession, Marital status, Education, Personal loan, House loan, Partner working.

1.2 Take a critical look at the data and do a preliminary analysis of the variables. Do a quality check of the data so that the variables are consistent. Are there any discrepancies present in the data? If yes, perform preliminary treatment of data.

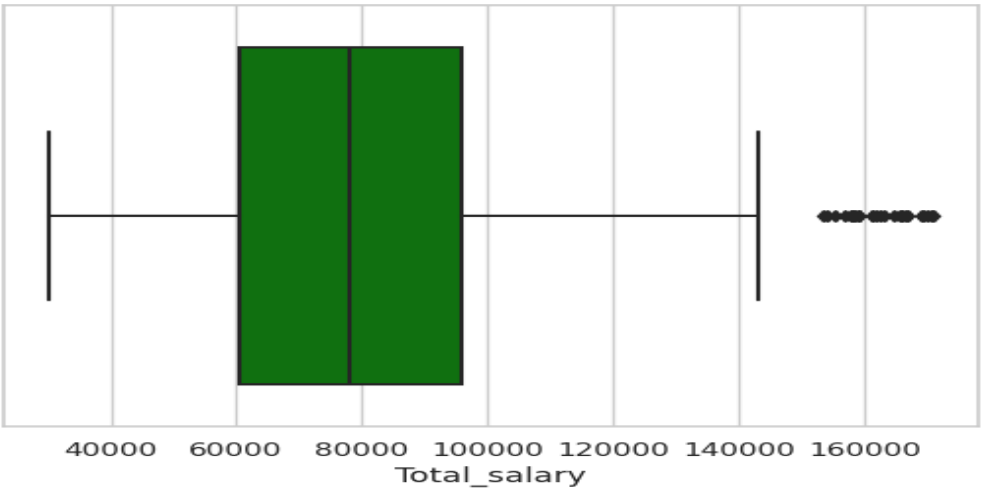
- **Missing Values:** It is observed that there are 53 null values in Gender and 106 null values in Partner salary variables

Filling the null values: Partner salary and Gender

- **Partner salary:** In the dataset it has been observed that there are 106 missing or null values. To fill the missing values a function was defined using the partner working variable. If the partner working is 'No' then the partner salary was replaced with the value '0'. If the partner working is 'yes' then the partner salary was replaced with the value output [partner salary = total salary – salary].
- **Gender:** Two discrepancies were found in the Gender variable data. [Female and Male] and 53 null values.

1. Unique values found [Femal, Femle]. Replaced the data with 'Female' if the data has the object type like Femal and Femle.
2. To fill the null values mode method was used as mode vale is Male replaced with the all the 53 missing values.

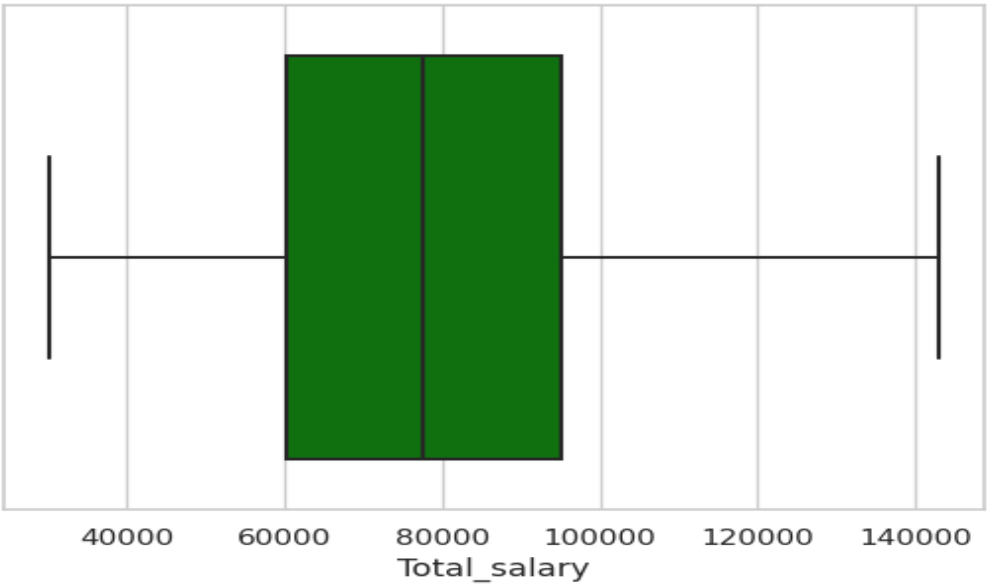
Outliers found in the variable Total Salary:



From the above boxplot its has been observed that there are some outliers in the Total Salary variable.

Percentile Method (IQR) is used to treat the outliers:

Using the IQR method and removed the outliers in the total salary and the updated boxplot as shown below.



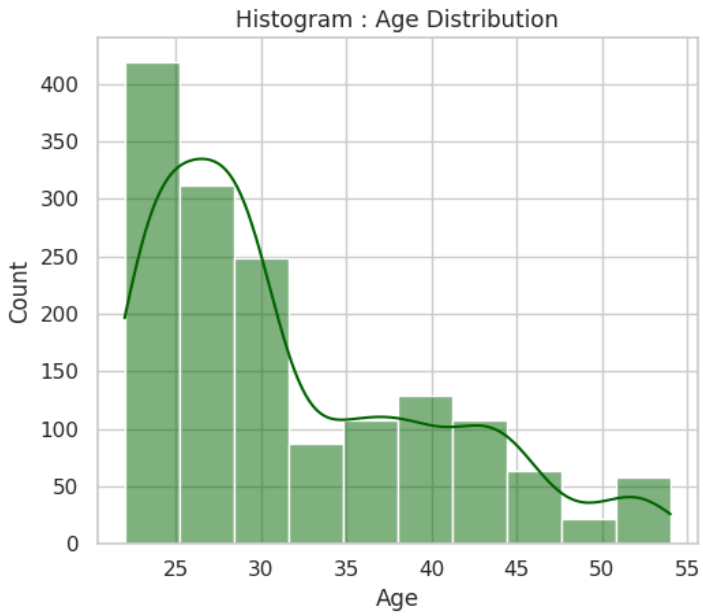
- No null values found in the dataset after cleaning the dataset using the isnull function.
- No duplicates found in the data set.

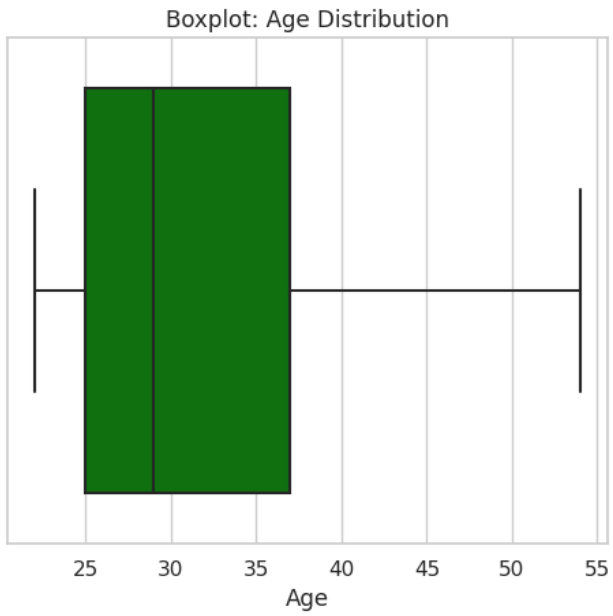
1.3 Explore all the features of the data separately by using appropriate visualizations and draw insights that can be utilized by the business.

Description - Numerical data:

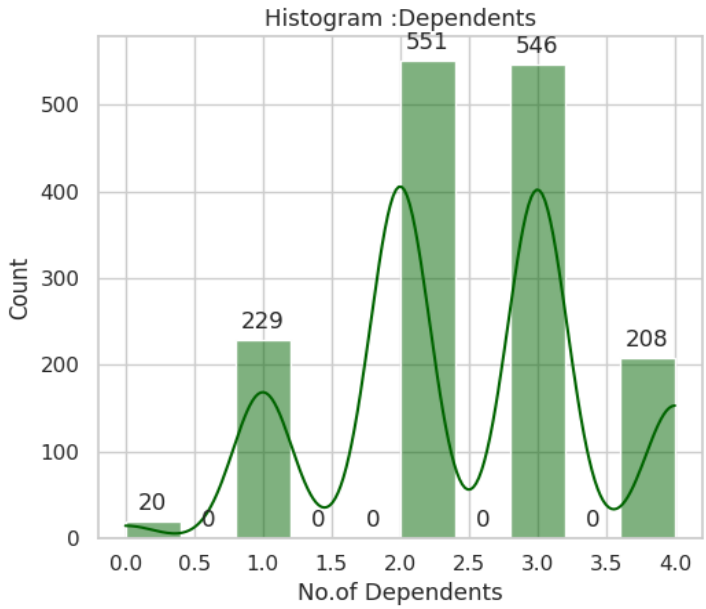
	count	mean	std	min	25%	50%	75%	max
Age	1554.0	31.624839	8.174540	22.0	25.0	29.0	37.0	54.0
No_of_Dependents	1554.0	2.445946	0.941960	0.0	2.0	2.0	3.0	4.0
Salary	1554.0	59813.449163	14112.929888	30000.0	51800.0	59200.0	71275.0	90700.0
Partner_salary	1554.0	18375.804376	18704.753352	0.0	0.0	24900.0	35600.0	70800.0
Total_salary	1554.0	78189.253539	23292.262623	30000.0	60200.0	77500.0	95050.0	143100.0
Price	1554.0	35142.213642	13286.277066	18000.0	24000.0	31000.0	45750.0	70000.0

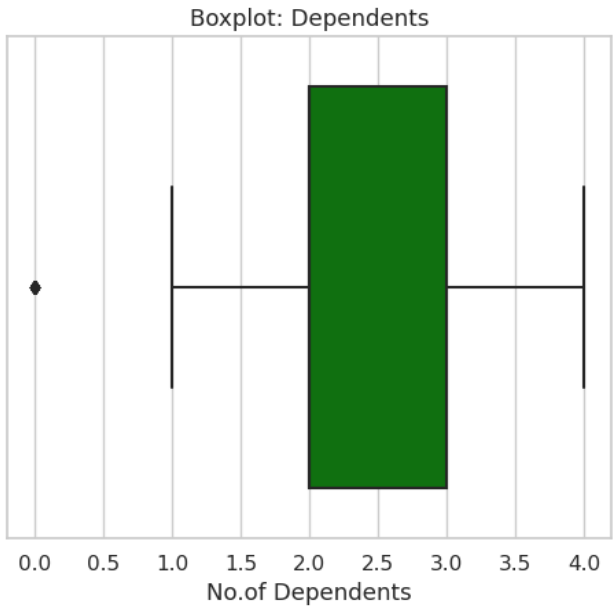
- **Age distribution:** From the below visualization it is indicating that a right-skewed distribution and ages of individuals, ranging from a min of 22 years to a max of 54 years. Majority of the ages fall within the range of 25 to 37, with a median age of 29. The distribution of ages is slightly right-skewed, as indicated by a positive skewness value of 0.92466037. The standard deviation of the ages is 8.17454.



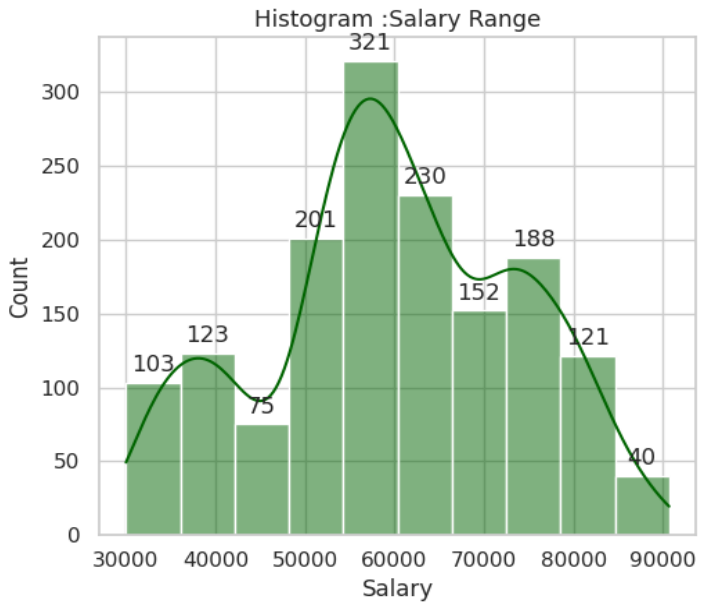


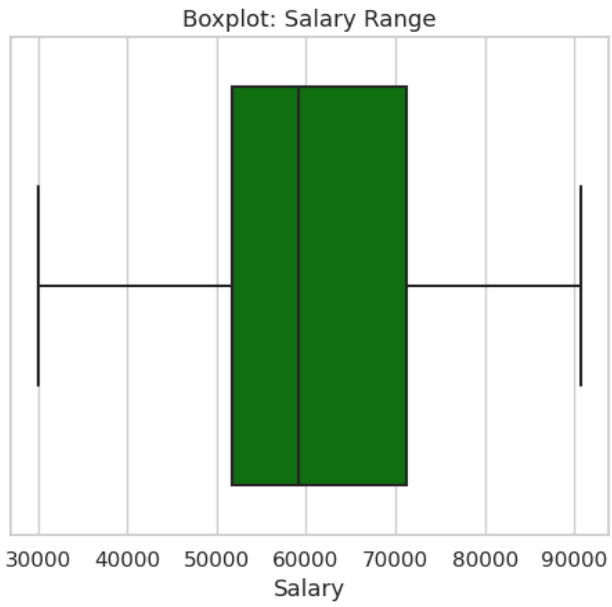
- **No. of Dependents:** From the graph it shows that a significant portion of the individuals in the study have 2 to 3 dependents, with 551 and 546 individuals falling into this category. Additionally, there are individuals with 1 and 4 dependents, with counts of 229 and 208 respectively and there are 20 individuals in the dataset who have no dependents.



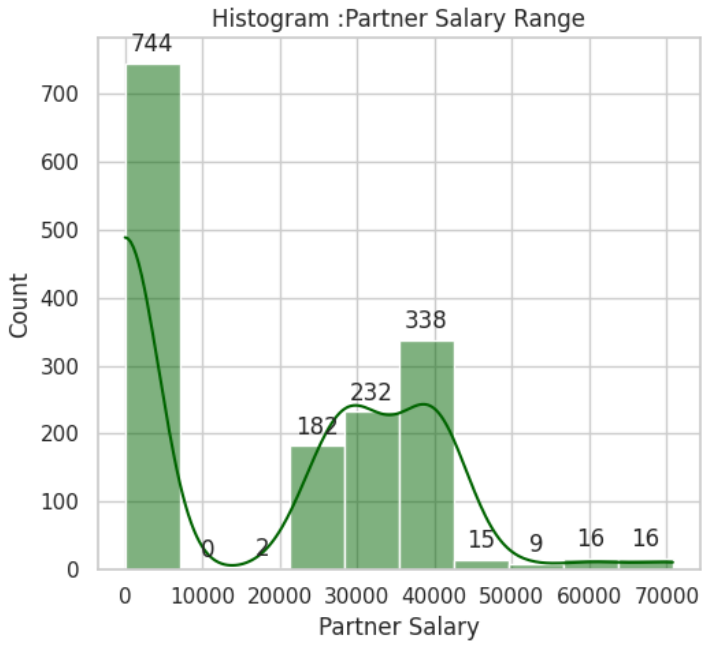


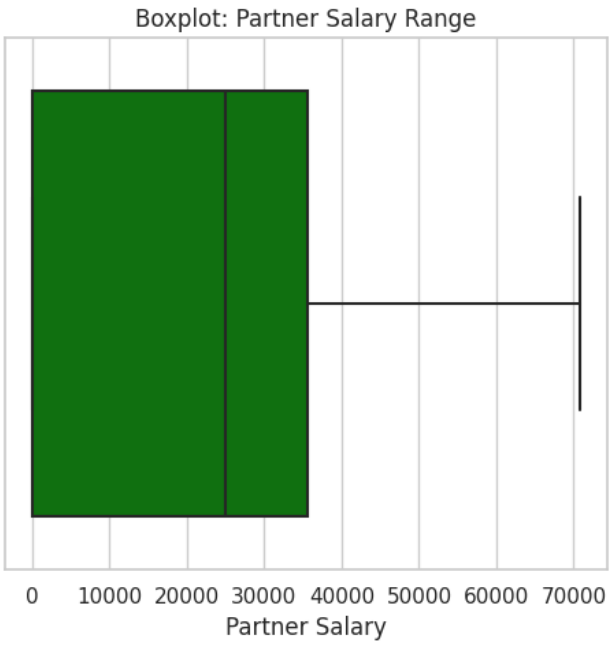
- **Salary:** From the below visualization it is indicating that the salary values ranging from a minimum of 30,000 to a maximum of 90,700. Majority of the salaries fall within the range of 51,800 to 71,275, with a median salary of 59,200. The salaries show a slightly negative skewness (-0.12915999), indicating a slightly left-skewed distribution. The standard deviation of the salaries is 14,112.93, the salaries exhibit a relatively normal distribution with a slight left-skewness, and there is a wide range of salary values, indicating different income levels among individuals in the study.



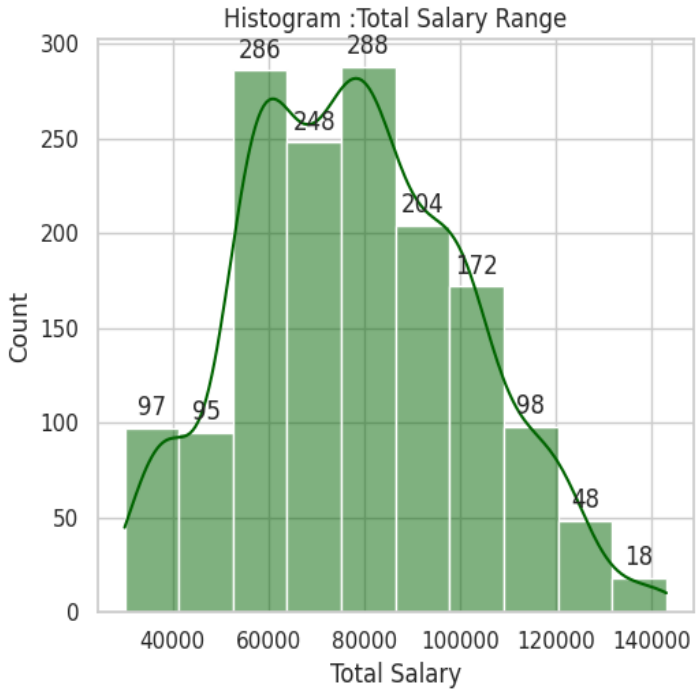


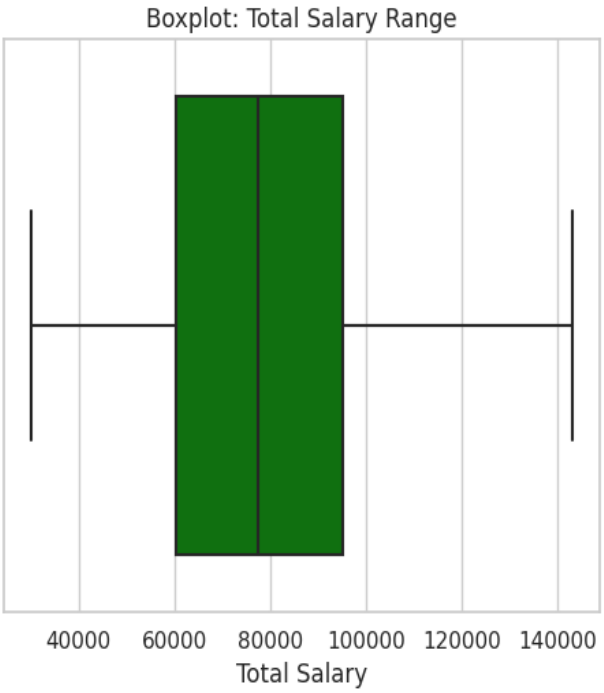
- **Partner salary:** Majority of the partners not earning any income and partner salaries, ranging from a minimum of 0 to a maximum of 70,800 and majority of partner salaries fall within the range of 0 to 35,600, with a median salary of 24,900. The distribution of partner salaries is slightly right-skewed, as indicated by a positive skewness value of 0.32463536. The standard deviation of partner salaries is 18,704, Overall, there is a wide range of partner salary values, with some individuals having no reported salary and others having higher salaries up to 70,800.



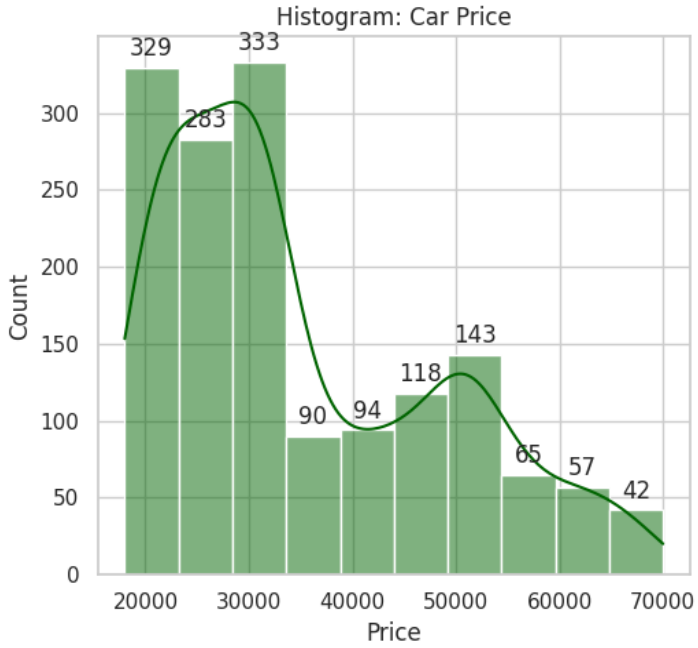


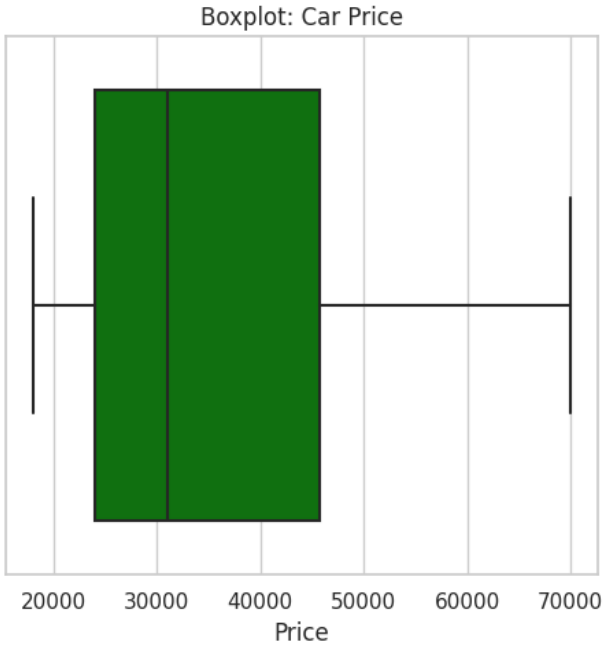
- **Total salary:** The total salaries, ranging from a minimum of 30,000 to a maximum of 143,100. Majority of the total salaries fall within the range of 60,200 to 95,050, with a median salary of 77,500. The distribution of total salaries is slightly right-skewed, as indicated by a positive skewness value of 0.21591336. The standard deviation of total salaries is 23,292 and there is a wide range of total salary values, with some individuals earning higher salaries up to 143,100.



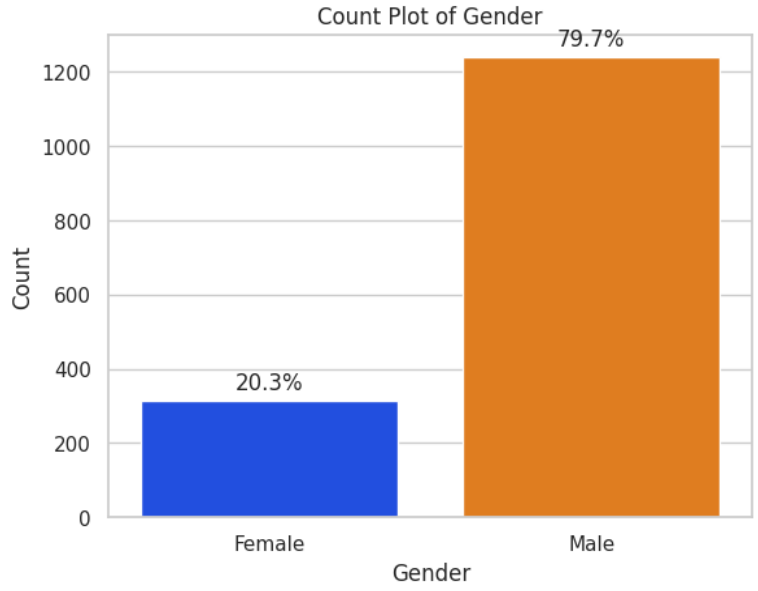


- **Price:** The prices of cars, ranging from a minimum of 18,000 to a maximum of 70,000. Majority of the prices fall within the range of 24,000 to 45,750, with a median price of 31,000. The distribution of prices is slightly right-skewed, as indicated by a positive skewness value of 0.76940062. The standard deviation of prices is 13,286.277 and there is a range of price values, with some cars priced higher up to 70,000.

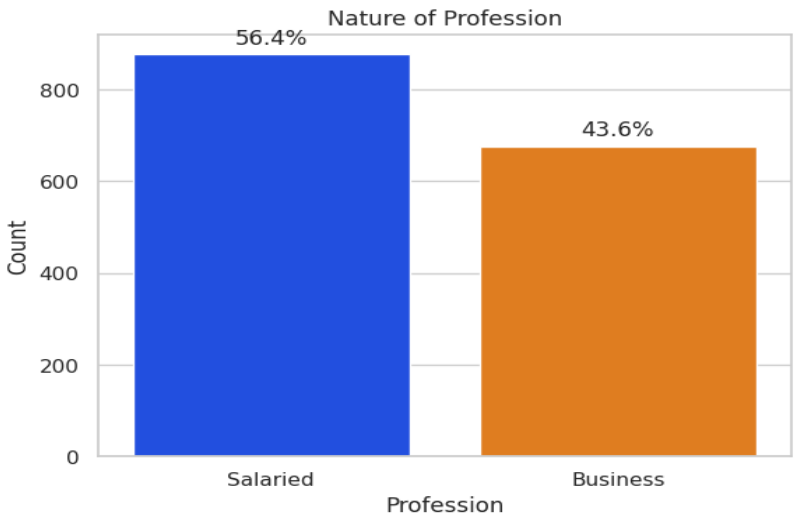




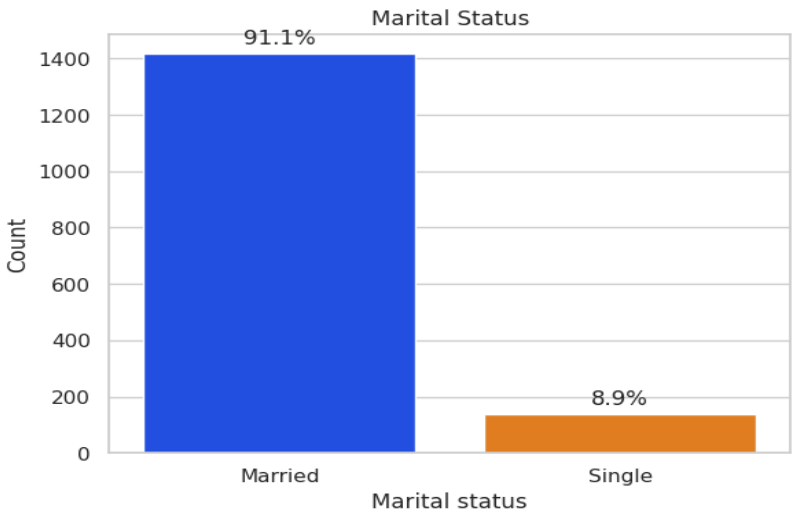
- **Gender:** It is observed that the sample consists of 80% male and 20 % of female individuals



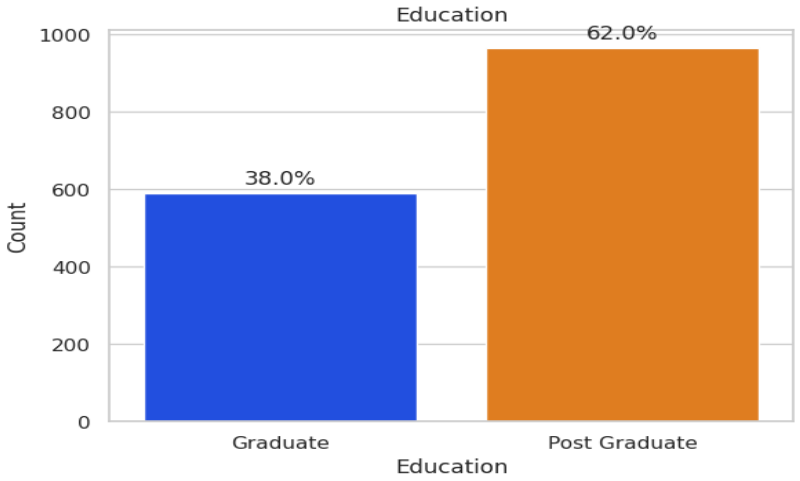
- **Profession:** It is observed in the sample that 56% of the individual are Salaried and 44% of the individuals having their own business.



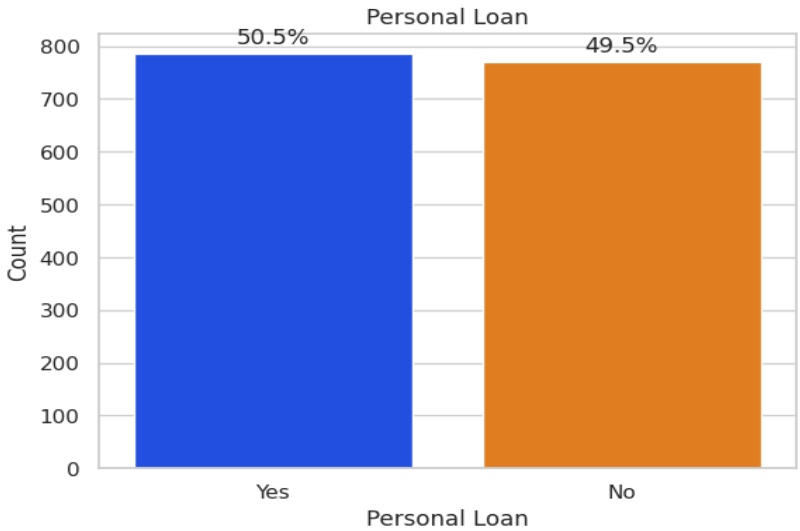
- **Marital status:** It is observed that 91% of the individuals are married and only 9% of individuals are single.



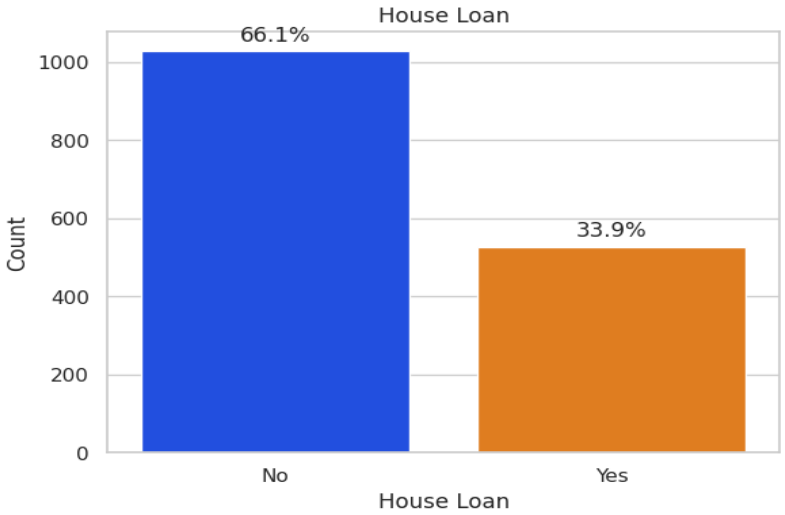
- **Education:** It is observed that 62% of the individuals completed their post-graduation and 38% only completed their graduation.



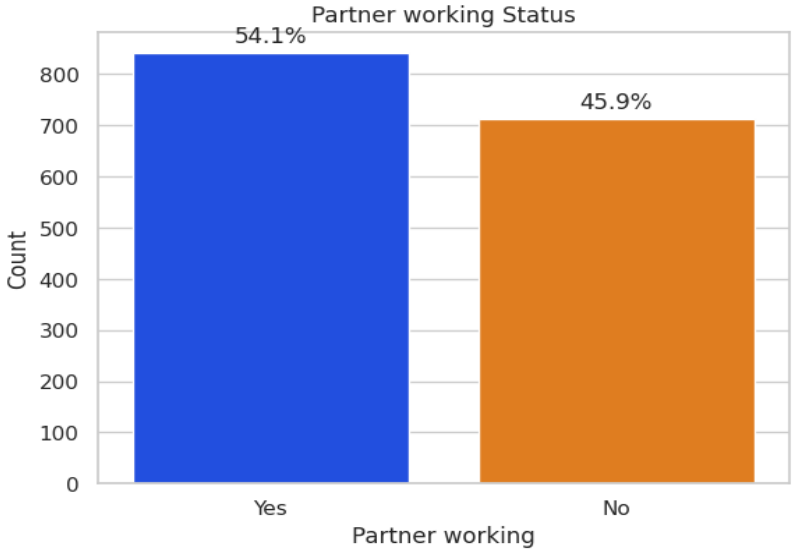
- **Personal loan:** It is observed that 50.5 % of the individuals took the personal loans and 49.5% of the individuals did not took any personal loans



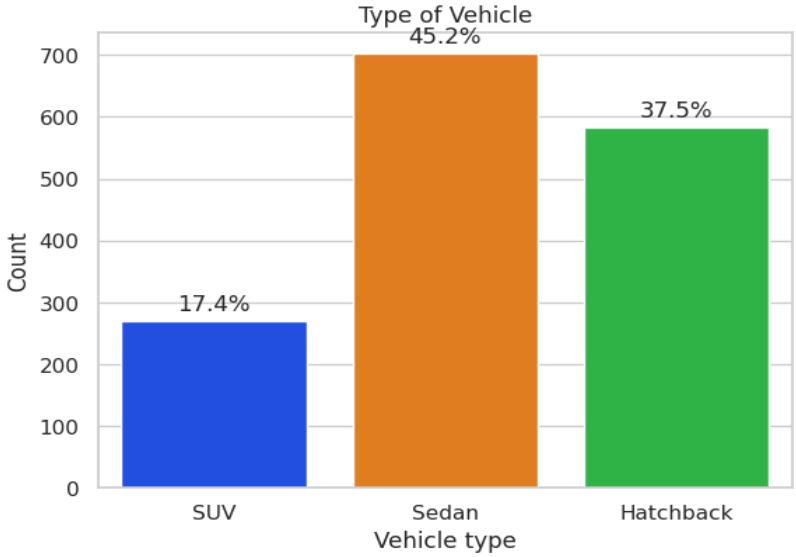
- **House loan:** It is observed that 66 % of the individuals took the housing loan and 34% of the individuals did not took any housing loan.



- **Partner working:** It is observed that 54% of the partners are working and 46% of the partners are not working.



- **Make:** It is observed that most of the individuals owning Sedan (45.2%) followed by Hatchback (37.5%) and the least owning vehicle is SUV (17.4%)



1.4 Understanding the relationships among the variables in the dataset is crucial for every analytical project. Perform analysis on the data fields to gain deeper insights. Comment on your understanding of the data.

- **Correlation between numerical values:**

```
Correlation Matrix:
                      Age  No_of_Dependents    Salary  Partner_salary  \
Age              1.000000         -0.227292  0.584401        0.052150
No_of_Dependents -0.227292          1.000000 -0.065071        0.119469
Salary           0.584401         -0.065071  1.000000       -0.012336
Partner_salary   0.052150          0.119469 -0.012336        1.000000
Total_salary     0.395971          0.056512  0.596000        0.795571
Price            0.785936         -0.169662  0.363324        0.094804

                 Total_salary     Price
Age                  0.395971  0.785936
No_of_Dependents     0.056512 -0.169662
Salary               0.596000  0.363324
Partner_salary       0.795571  0.094804
Total_salary         1.000000  0.296273
Price                0.296273  1.000000
```

Age: Positive correlation with Salary (0.584): As age increases, the salary tends to increase.

Positive correlation with Total salary (0.396): Older individuals tend to have higher total salaries.

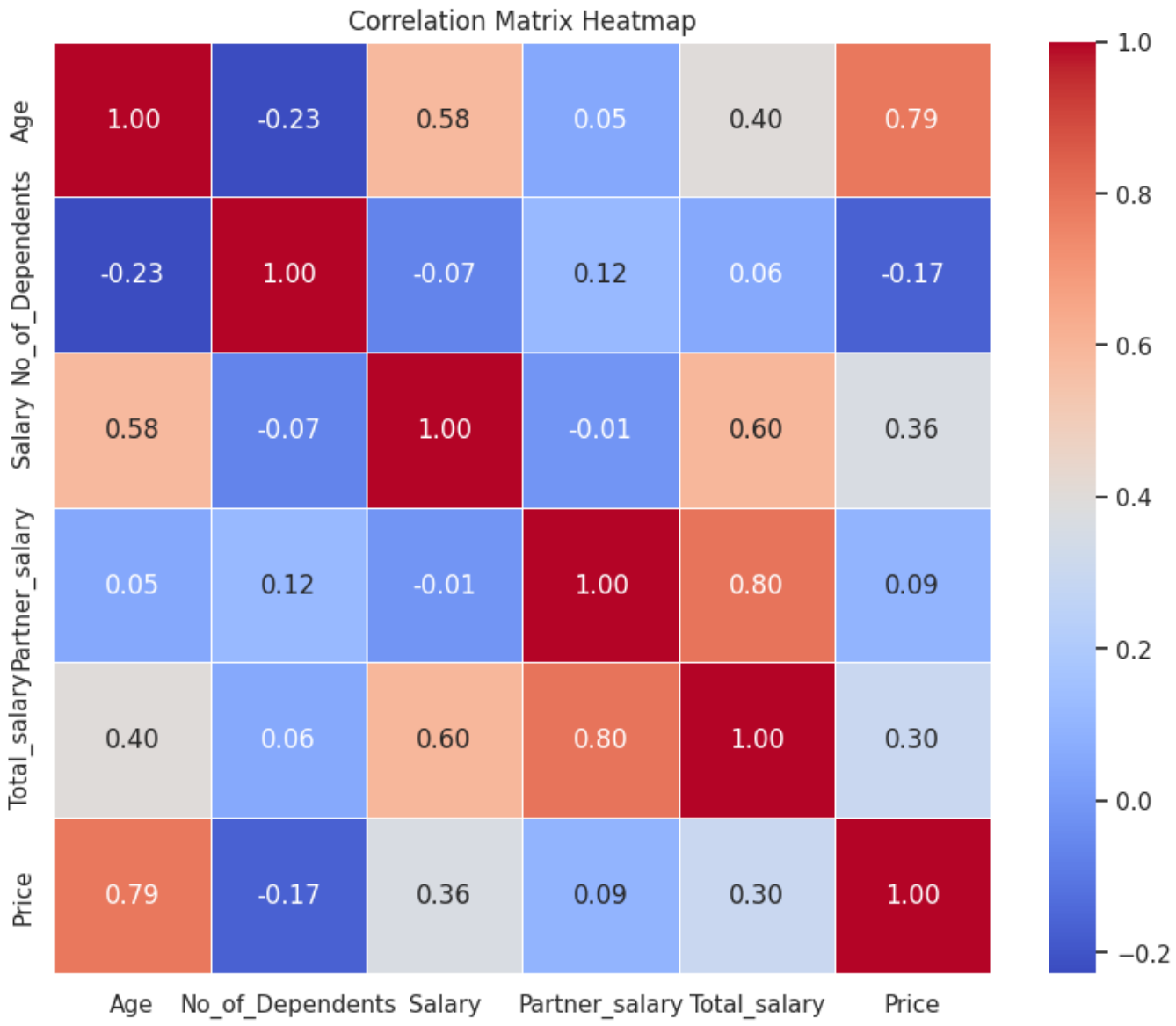
No of Dependents: Negative correlation with Age (-0.227): As the number of dependents increases, the age tends to be lower. Negative correlation with Price (-0.170): More dependents are associated with lower prices.

Salary: Positive correlation with Age (0.584): Salary tends to increase with age. Positive correlation with total salary (0.596): Salary contributes to the total salary.

Partner_salary: Positive correlation with total salary (0.796): Partner's salary strongly contributes to the total salary.

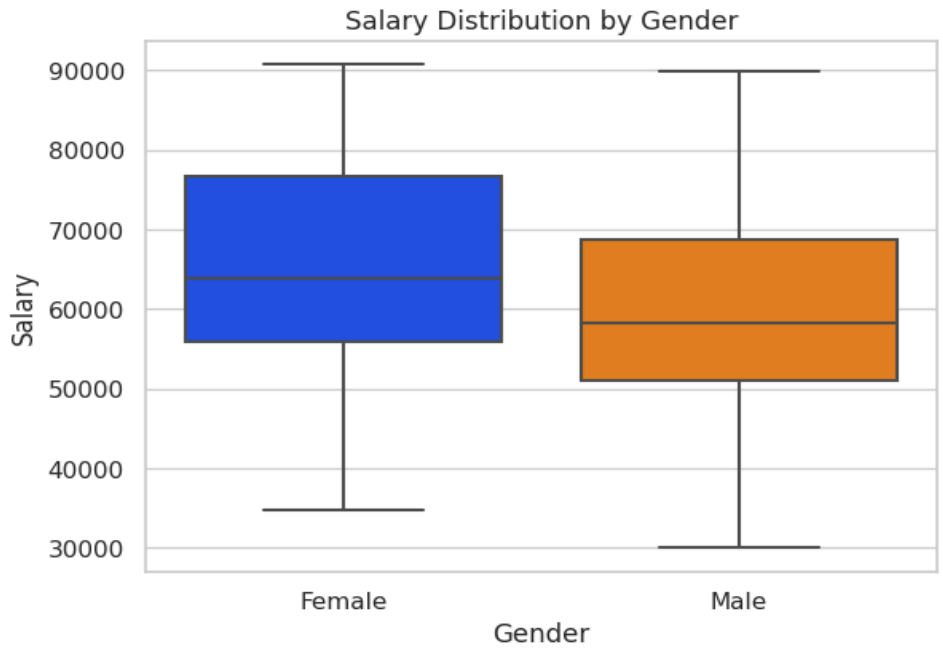
Total salary: Positive correlation with Age (0.396): Total salary tends to increase with age. Positive correlation with Salary (0.596): Total salary is positively influenced by individual salary. Positive correlation with Partner_salary (0.796): Total salary is strongly influenced by partner's salary. Positive correlation with Price (0.296): Higher total salary is associated with higher prices.

Correlation Heatmap:



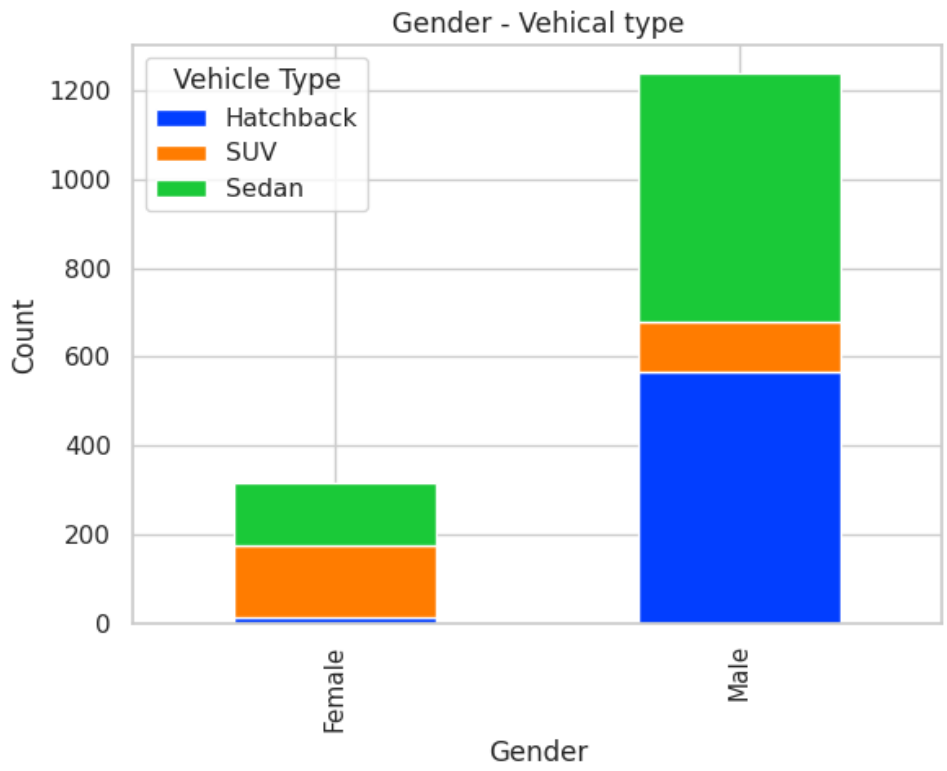
The correlation matrix gives insights and relationships between the variables. For example, age has a positive influence on salary, total salary, and price. The number of dependents has a negative impact on age and price. Salary and partner's salary contribute to the total salary.

Salary distribution – Gender:



From the above graph the Female individuals are earning higher income then the Male individuals.

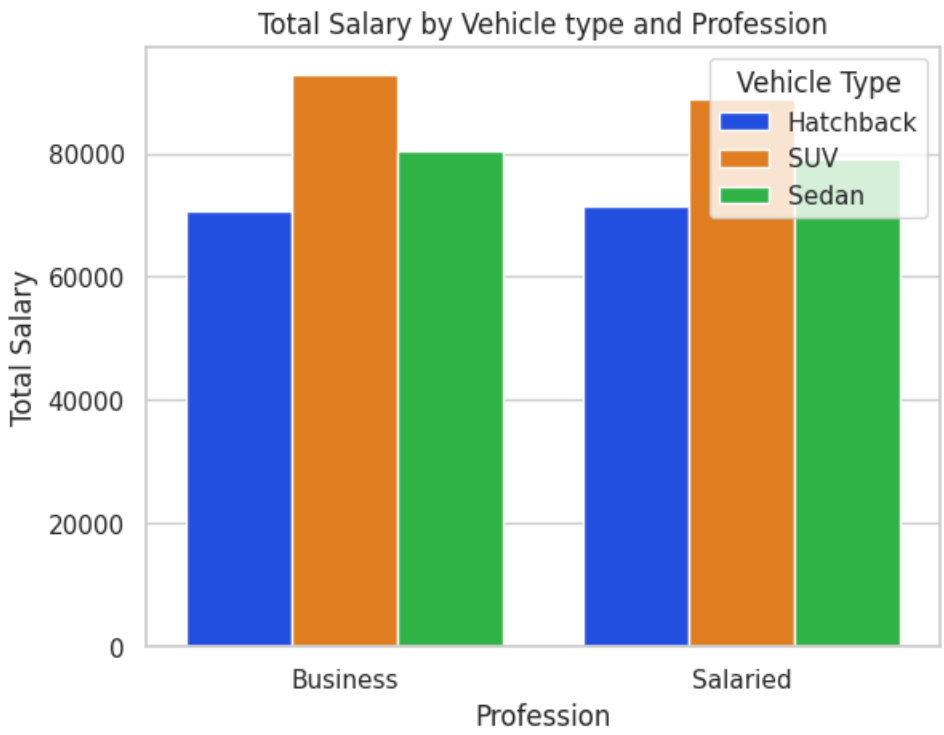
Gender and Vehicle type:



From the above graph it is observed that Male individuals have majority of Hatchback and Sedan also female individuals have SUV and Sedan.

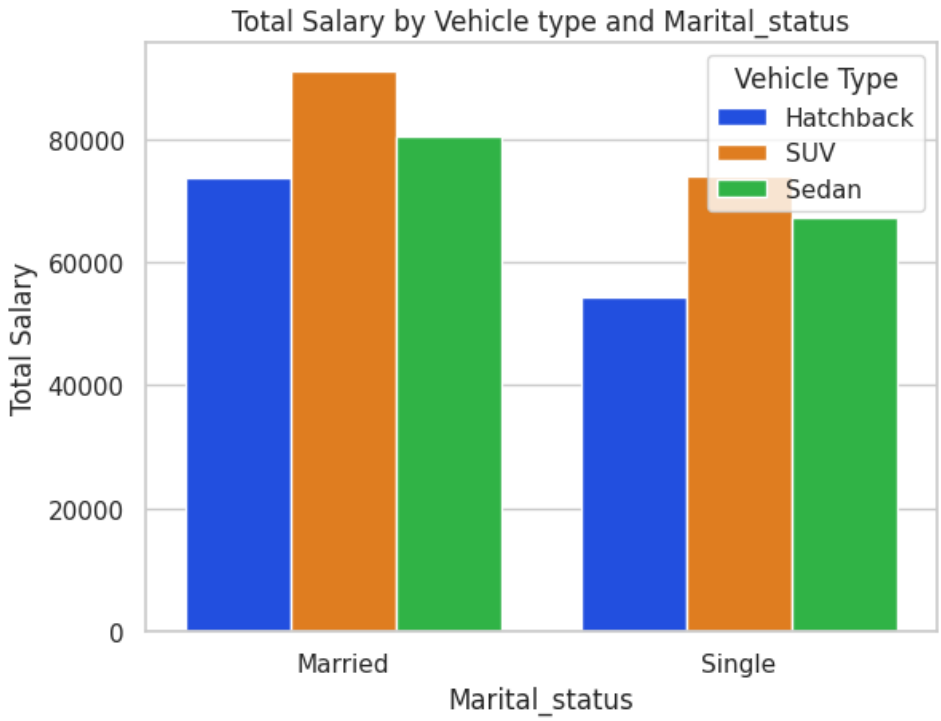
Vehicle choice based on total salary and profession:

In the below graph it is observed that in both business and salaried individuals chooses SUV followed by Sedan.



Vehicle choice based on total salary and Marital status:

In the below graph it is observed that in both Married and single individuals chooses SUV followed by Sedan.



1.5 Employees working on the existing marketing campaign have made the following remarks. Based on the data and your analysis state whether you agree or disagree with their observations. Justify your answer Based on the data available.

1.5.2 Steve Roger says "Men prefer SUV by a large margin, compared to the women"

As per the analysis, Men do not prefer SUV by a large margin, compared to women

1.5.3 Ned Stark believes that a salaried person is more likely to buy a Sedan.

As per the analysis, Salaried individuals are not more likely to buy a Sedan

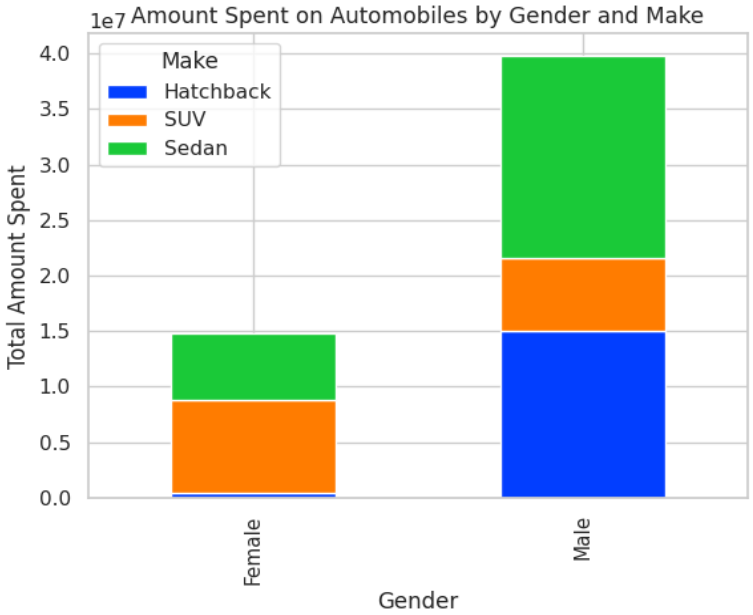
1.5.4 Sheldon Cooper does not believe any of them; he claims that a salaried male is an easier target for a SUV sale over a Sedan Sale.

As per the analysis, Salaried males are an easier target for a Sedan sale over a SUV

1.6 From the given data, comment on the amount spent on purchasing automobiles across the following categories. Comment on how a business can utilize the results from this exercise. Give justification along with presenting metrics/charts used for arriving at the conclusions. Give justification along with presenting metrics/charts used for arriving at the conclusions.

1.6.2 Gender

Gender	Make	
Female	Hatchback	412000
	SUV	8372000
	Sedan	6031000
Male	Hatchback	14996000
	SUV	6539000
	Sedan	18261000



Females have spent a total of 8,372,000 on SUVs, 6,031,000 on sedans, and 412,000 on hatchbacks. Males have spent a total of 6,539,000 on SUVs, 18,261,000 on sedans, and 14,996,000 on hatchbacks.

Targeted Marketing: The analysis gives insights on gender specific choices for different vehicle makes. For example, females show a higher preference for SUVs and sedans, while males show a higher preference for sedans and hatchbacks. The data driven decision making can be used for business purposes in their marketing strategies and advertisements to target specific gender-vehicle make combinations. For instance, advertising SUVs to females and sedans to males may yield better results.

Product Offering: The spending patterns by gender and vehicle make can guide the business in determining their product offerings. If there is a significant gender preference for a particular vehicle

make, the business can consider expanding their inventory or negotiating better deals with manufacturers to ensure the availability of popular vehicle makes for their target audience.

1.6.3 Personal loan

Make	Hatchback	SUV	Sedan
Personal_loan			
No	7765000	9134000	10852000
Yes	7643000	5777000	13440000

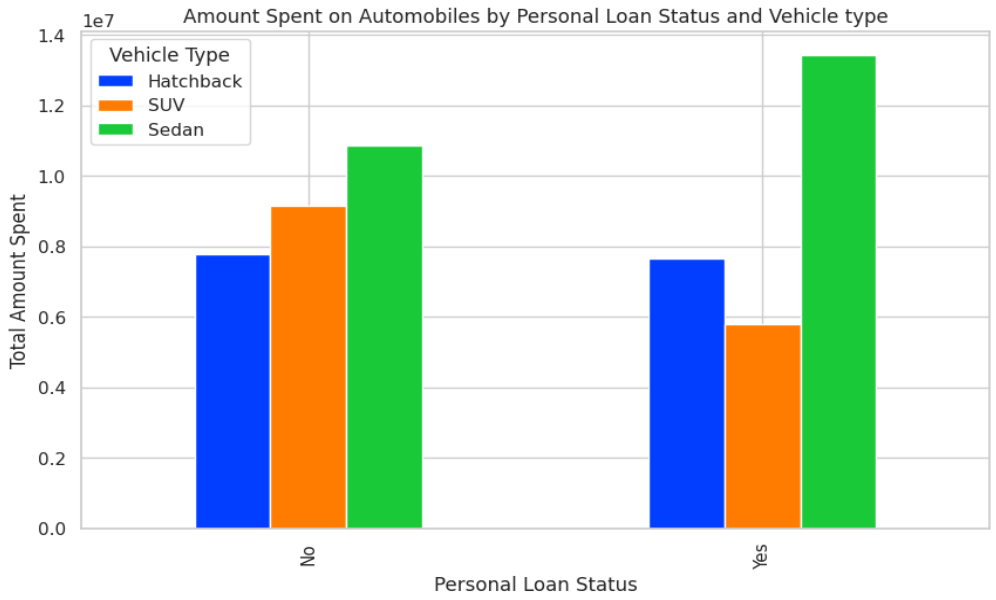
Based on the analysis of the amount spent on automobiles by personal loan status and vehicle make, the following insights can be derived:

Spending by Personal Loan: Customers with no personal loan have spent a total of 27,410,000 across all vehicle makes. Customers with a personal loan have spent a total of 27,350,000 across all vehicle makes. The spending is comparable between the two groups, indicating that personal loan status alone does not significantly impact the amount spent on automobiles.

Spending by Vehicle Make: Sedan is the most popular vehicle make among customers with no personal loan, with a total spending of 10,852,000. This is followed by SUV (9,134,000) and Hatchback (7,765,000). Among customers with a personal loan, Sedan also remains the most popular make, with a significantly higher spending of 13,440,000. This is followed by Hatchback (7,643,000) and SUV (5,777,000).

Targeted Marketing: Since Sedan is the most popular make across both customer segments, the business can focus its marketing efforts on promoting Sedan models to attract potential customers. Additionally, the business can target customers with personal loans specifically for Sedan models, as they tend to spend a higher amount.

Loan Offerings: Based on the spending patterns, the business can design personalized loan offerings for customers interested in Sedans, as they show a higher inclination towards purchasing such vehicles. By providing attractive loan options for Sedan purchases, the business can further incentivize customers with personal loans to make a purchase.



The analysis was performed using a grouped bar chart that visualized the total amount spent on automobiles by personal loan status and vehicle make. This chart allowed for a clear comparison between different makes and loan statuses, enabling the identification of trends and patterns in customer spending behaviour. The summarized metrics provided a quantitative understanding of the spending distribution, which informed the strategic recommendations for the business.

1.7 From the current data set comment if having a working partner leads to the purchase of a higher-priced car.

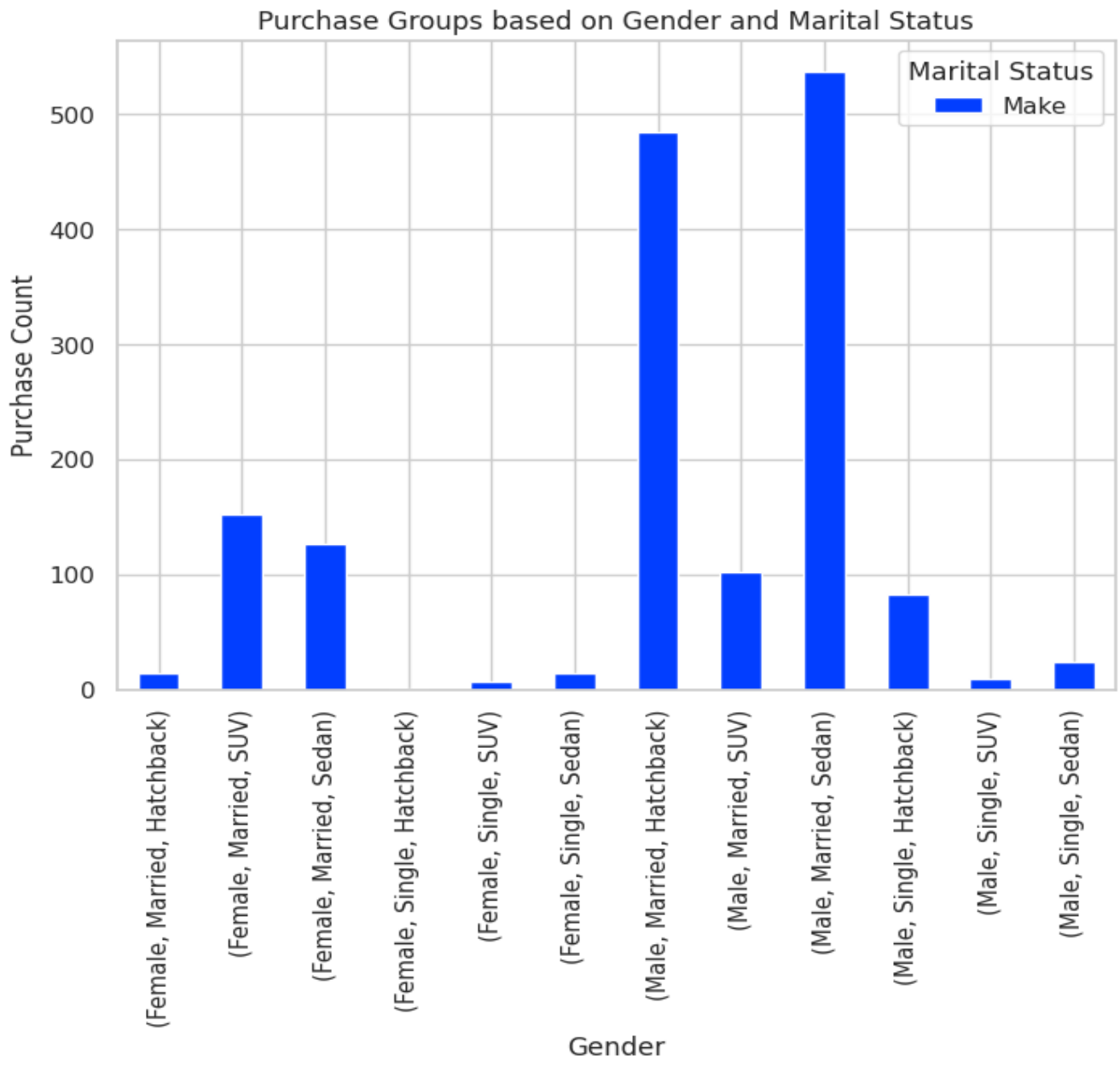
Based on the analysis of the data, it can be concluded that having a working partner does not lead to the purchase of a higher-priced car. The average car price for individuals without a working partner is $36,000, while for those with a working partner, it is approximately $34,415. Therefore, there is no significant difference in car prices based on the presence of a working partner.

This information can be utilized by businesses to develop targeted marketing strategies that do not solely focus on individuals with working partners as potential customers for higher-priced cars.

1.8 The main objective of this analysis is to devise an improved marketing strategy to send targeted information to different groups of potential buyers present in the data. For the current analysis use the Gender and Marital status - fields to arrive at groups with similar purchase history.

```
Gender  Marital_status  Make
Female  Married         Hatchback     14
                        SUV          152
                        Sedan        127
        Single          Hatchback      1
                        SUV            7
                        Sedan         14
Male    Married         Hatchback    484
                        SUV          102
                        Sedan        537
        Single          Hatchback     83
                        SUV            9
                        Sedan         24
```

- Based on the analysis of the data, majority of purchases are made by married individuals, with a higher preference for Sedan cars among both males and females.
- Single individuals have a relatively lower purchase count across all car types compared to married individuals.
- Females show a preference for SUV cars, followed by Sedan and Hatchback, while males have a higher purchase count for Sedan cars, followed by Hatchback and SUV.
- Targeted marketing efforts can be focused on married individuals, particularly males, who have a higher purchase count for Sedan cars.
- Special attention can be given to married females who show a significant preference for SUV cars, providing them with tailored information and offers related to SUV models.



By understanding the purchase behavior of different groups based on Gender and Marital Status, the business can customize its marketing strategy to effectively reach potential buyers and offer them relevant information and incentives based on their preferences.

1.9 Analyze the dataset and list down the top 5 important variables, along with the business justifications

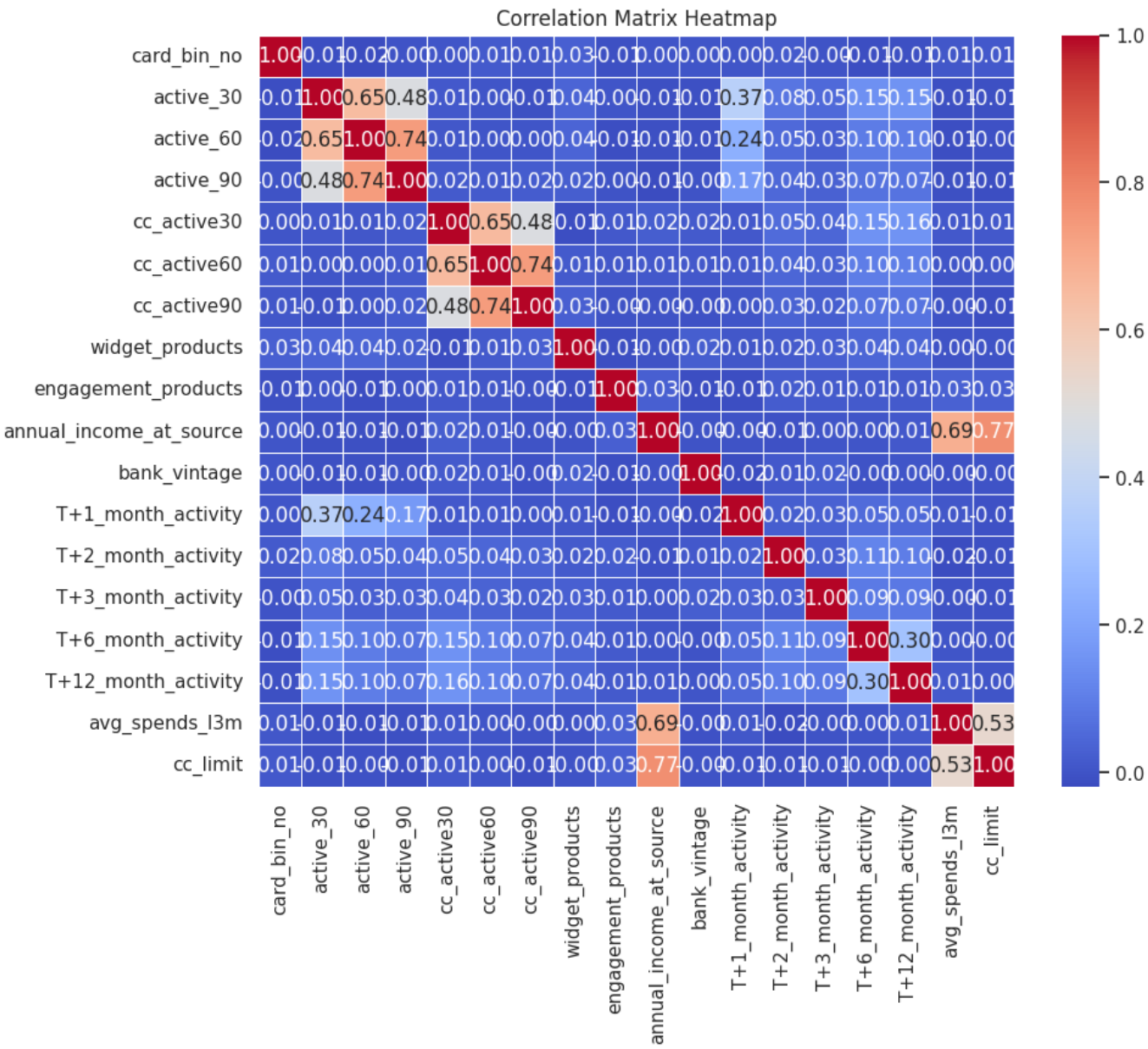
1. Shape of the dataset: (8448, 28)

Dataset Information:

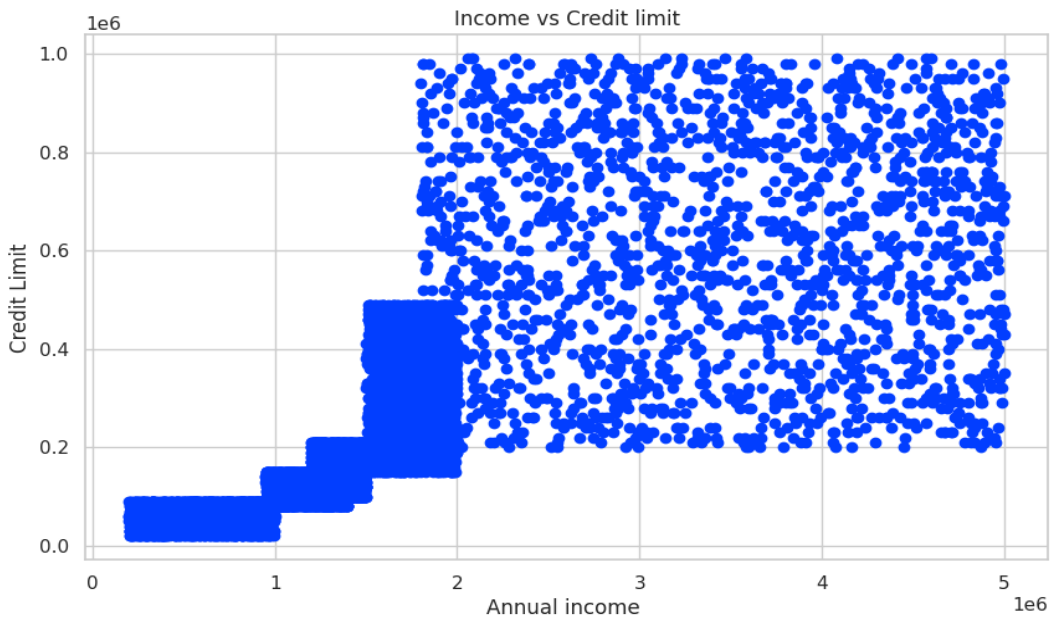
```
RangeIndex: 8448 entries, 0 to 8447
Data columns (total 28 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   userid                    8448 non-null   int64
 1   card_no                   8448 non-null   object
 2   card_bin_no               8448 non-null   int64
 3   Issuer                    8448 non-null   object
 4   card_type                 8448 non-null   object
 5   card_source_date          8448 non-null   datetime64[ns]
 6   high_networth             8448 non-null   object
 7   active_30                 8448 non-null   int64
 8   active_60                 8448 non-null   int64
 9   active_90                 8448 non-null   int64
 10  cc_active30               8448 non-null   int64
 11  cc_active60               8448 non-null   int64
 12  cc_active90               8448 non-null   int64
 13  hotlist_flag              8448 non-null   object
 14  widget_products           8448 non-null   int64
 15  engagement_products       8448 non-null   int64
 16  annual_income_at_source   8448 non-null   int64
 17  other_bank_cc_holding     8448 non-null   object
 18  bank_vintage              8448 non-null   int64
 19  T+1_month_activity        8448 non-null   int64
 20  T+2_month_activity        8448 non-null   int64
 21  T+3_month_activity        8448 non-null   int64
 22  T+6_month_activity        8448 non-null   int64
 23  T+12_month_activity       8448 non-null   int64
 24  Transactor_revolver       8410 non-null   object
 25  avg_spends_l3m            8448 non-null   int64
 26  Occupation_at_source      8448 non-null   object
 27  cc_limit                  8448 non-null   int64
dtypes: datetime64[ns](1), int64(19), object(8)
memory usage: 1.8+ MB
```

Dataset Description:

	count	mean	std	min	25%	50%	75%	max
card_bin_no	8410.0	4.367706e+05	3.051686e+04	376916.0	426241.0	437551.0	438439.0	524178.0
active_30	8410.0	2.925089e-01	4.549418e-01	0.0	0.0	0.0	1.0	1.0
active_60	8410.0	4.947681e-01	5.000024e-01	0.0	0.0	0.0	1.0	1.0
active_90	8410.0	6.424495e-01	4.793073e-01	0.0	0.0	1.0	1.0	1.0
cc_active30	8410.0	2.839477e-01	4.509385e-01	0.0	0.0	0.0	1.0	1.0
cc_active60	8410.0	4.840666e-01	4.997758e-01	0.0	0.0	0.0	1.0	1.0
cc_active90	8410.0	6.318668e-01	4.823264e-01	0.0	0.0	1.0	1.0	1.0
widget_products	8410.0	3.625327e+00	2.272034e+00	0.0	2.0	4.0	6.0	7.0
engagement_products	8410.0	4.006778e+00	2.567130e+00	0.0	2.0	4.0	6.0	8.0
annual_income_at_source	8410.0	1.674374e+06	1.064592e+06	200095.0	1061494.5	1371687.0	1881414.5	4999508.0
bank_vintage	8410.0	3.315779e+01	1.587166e+01	6.0	19.0	33.0	47.0	60.0
T+1_month_activity	8410.0	1.117717e-01	3.151041e-01	0.0	0.0	0.0	0.0	1.0
T+2_month_activity	8410.0	4.815696e-02	2.141105e-01	0.0	0.0	0.0	0.0	1.0
T+3_month_activity	8410.0	8.073722e-02	2.724473e-01	0.0	0.0	0.0	0.0	1.0
T+6_month_activity	8410.0	8.917955e-03	9.401849e-02	0.0	0.0	0.0	0.0	1.0
T+12_month_activity	8410.0	9.512485e-03	9.707275e-02	0.0	0.0	0.0	0.0	1.0
avg_spends_l3m	8410.0	4.975115e+04	4.622905e+04	21.0	17260.0	38094.0	66204.5	289292.0
cc_limit	8410.0	2.515850e+05	2.290665e+05	20000.0	90000.0	150000.0	350000.0	990000.0



2. Annual income at source and credit limit:

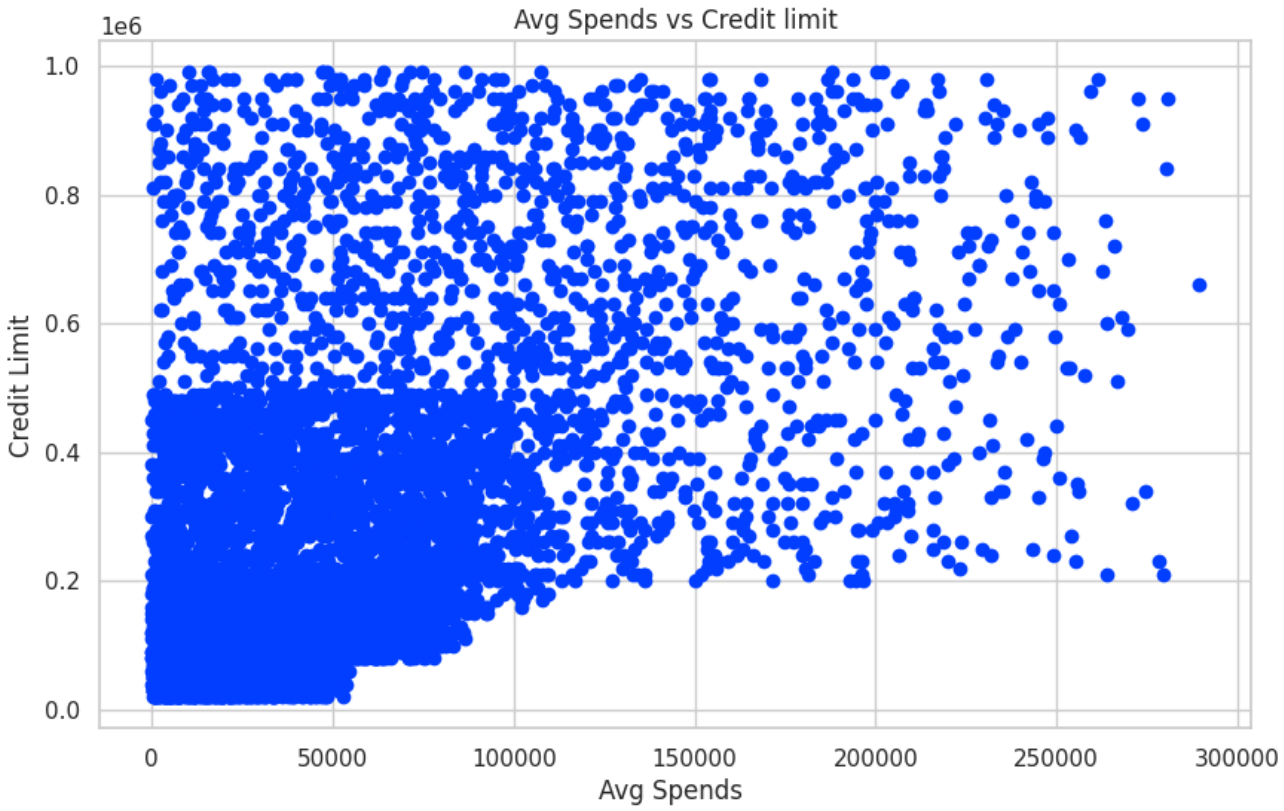


The correlation between annual_income_at_source and cc_limit is 0.772. This positive correlation suggests that there is a strong relationship between these two variables.

In a business context, this correlation implies that individuals with higher annual incomes tend to have higher credit limits. This could be due to the fact that higher income individuals are generally considered less risky by lenders, making them eligible for larger credit limits. On the other side, individuals with lower incomes may have lower credit limits as they might be perceived as having a higher default risk.

Understanding the correlation between annual income and credit limit can be valuable for businesses in various ways GODIGT can utilize this information to assess creditworthiness and determine appropriate credit limits for their customers which help businesses in targeting specific customer segments based on income levels and credit requirements, enabling them to tailor their products and services more effectively.

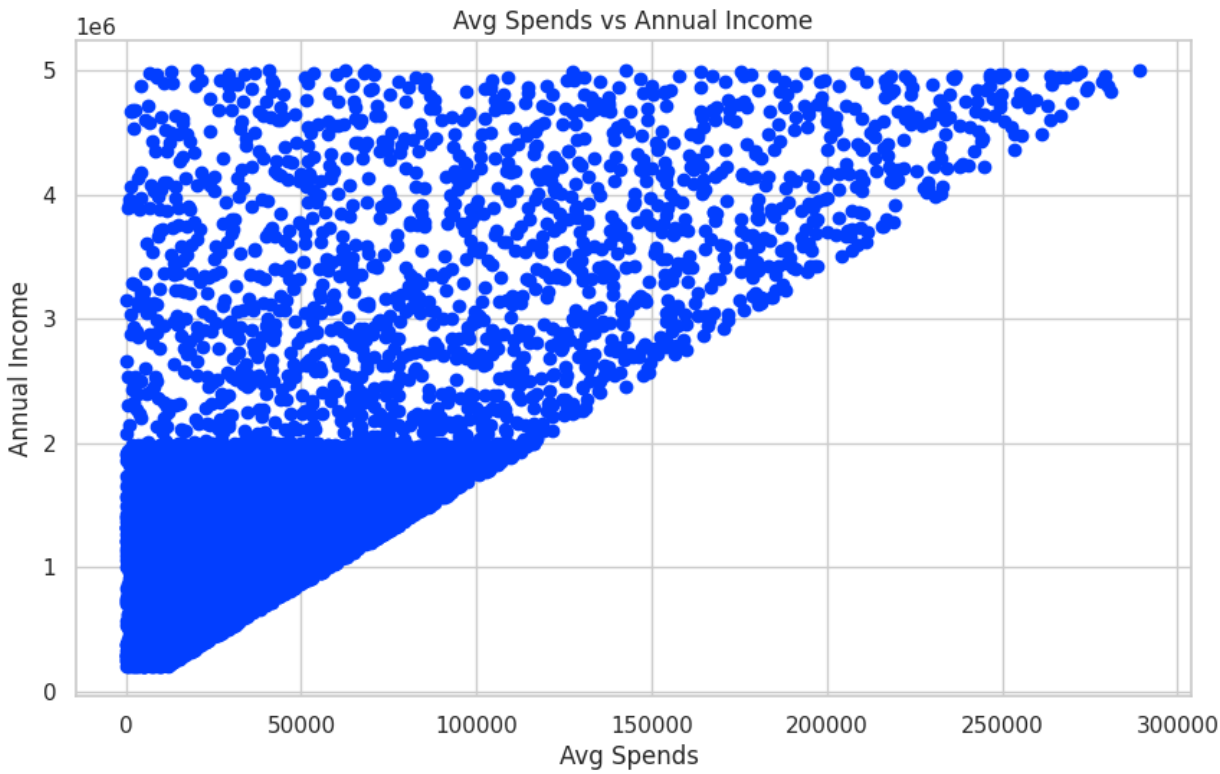
3. **Aug Spends and Credit Limit:**



The correlation of 0.5305 suggests a moderate positive correlation between the variables average spending and creditcard limit in the dataset which means that there is a tendency for an increase in average spending over the last 3 months to be associated with a higher credit limit.

From a business perspective, this correlation could indicate that customers who spend more on average in the last 3 months are more likely to have higher credit limits. This information can be valuable for GODIGT (credit card issuers) to understand the relationship between spending behavior and credit limits when making decisions such as credit limit adjustments, risk assessments, or offering tailored financial products to customers.

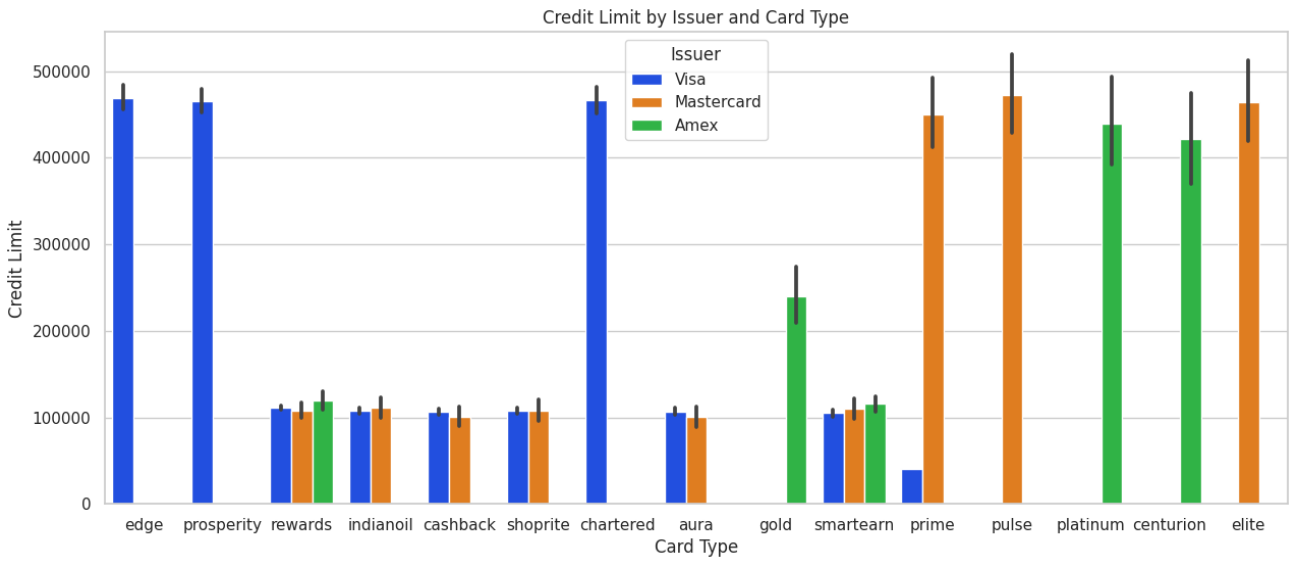
4. **Average spending and Annual Income :**



The correlation of 0.6881 indicates a moderate positive correlation between the variables average spending and Annual income at source in the dataset and the correlation suggests that there is a tendency for individuals with higher average spending over the last 3 months to have higher annual incomes

From a business perspective, this correlation can provide insights into the relationship between spending behavior and annual income. It implies that customers with higher incomes are more likely to have higher average spending levels. This information can be useful for businesses in various ways, such as targeted marketing strategies, product offerings, and financial planning services tailored to customers with different income levels.

5. **Credit Limit by Issuer and Card Type**



The above chart represents average credit limit for different card types offered by various issuers. The credit limit plays a crucial role in determining a cardholder's purchasing power and financial flexibility. Understanding the average credit limit for different card types and issuers can provide valuable insights into their offerings and target customer segments.

Among the issuers, Amex offers a diverse range of card types, with platinum having the highest average credit limit of approximately 439,661.02 units, followed by centurion and gold. Mastercard, on the other hand, offers a wide selection of card types, including elite, pulse, and prime, with elite having the highest average credit limit of around 463,854

Visa, another prominent issuer, offers various card types as well. Chartered and edge are the top two card types with the highest average credit limits, approximately 466,844.98 and 468,523.08 units, respectively. Other notable card types include prosperity, rewards, and shoprite.

Understanding the average credit limits provided by different issuers and card types can help individuals make informed decisions when selecting a credit card that aligns with their financial goals and purchasing habits. It can also assist GODIGT bank in analyzing market trends, evaluating the competitiveness of their card offerings, and developing targeted marketing strategies and this information can be useful for GODIGT bank in assessing the creditworthiness of applicants and managing credit risk based on the average credit limits associated with different card types and issuers. By leveraging this data, GODIGT bank can make data-driven decisions to optimize their credit card offerings and provide tailored solutions to meet the diverse needs of their customers.